Fault Atlas

Cryptography

```
❯ grep -B 2 -A 8 "if rnd == 9" model.py

        # Fault injection hook in Round 9 right before MixColumns
        if rnd == 9 and fault is not None:
            c, r, v = fault
            state[4 * c + r] ^= v

        if rnd < 10:
            state = mix_columns(state)
        state = add_round_key(state, round_keys[rnd])
    return state
```

I began by reading `model.py` and `data.json`. The challenge implemented AES-128 encryption with a fault injection hook triggered during Round 9 immediately before MixColumns.

The dataset provided 32 pairs of correct and faulty ciphertexts, allowing Differential Fault Analysis (DFA) against the final round state.

Because a single-byte fault in Round 9 diffuses into exactly four bytes of the ciphertext via MixColumns, I evaluated candidate Round-10 key bytes against the inverse S-Box differential equations. Inverting the AES key schedule recovered the original Round-0 master key.

Finally, I tested the recovered flag against the executable / service:

`python3 Fault_Atlas.py`

Round-10 key recovered via DFA. Key schedule inverted.

Flag Captured

`SecLeaf{f4ult5_d0nt_n33d_t0_b3_h0n35t}`

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